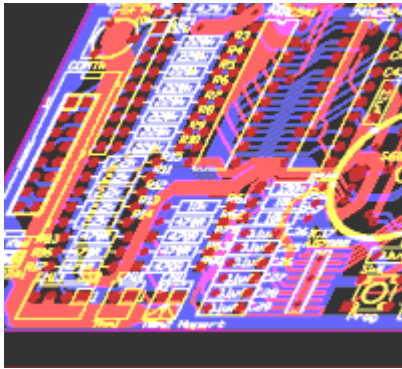


The Printed Circuit Board (PCB) Layout



The connections on the PCB should be identical to the circuit diagram, but while the circuit diagram is arranged to be readable, the PCB layout is arranged to be functional, so there is rarely any visible correlation between the circuit and the layout.

PCB layout can be performed manually (using CAD) or in combination with an Autorouter. The best results are usually still achieved using atleast some **manual** routing - simply because the design engineer has a

far better judgement of how to arrange circuitry. Surprisingly, many autorouted boards are often completely illogical in their track routing - the program has optimised the connections, and sacrificed any small amount of order that may have been put in place by manual routing. Generally autorouted boards are slightly harder for a technician to repair or debug, for this reason. Historically, PCBs used to be laid out by drawing or using stick on paper shapes on mylar film, - that really *WAS* manual routing!

The CAD PCB layout consists of several layers, for illustration purposes the holes, outline and component identification layers can be combined into one diagram. When we produce PCBs for clients, we use actual size checkplots during the design process. A PCB will usually have mounting holes and may possibly have cutouts, by cutting an actual size checkplot and placing it in the enclosure, you can see how it will be positioned in relation to other parts. We can also place components up against the pad markings as a quick double check of sizing.

The secret to creativity is knowing how to hide your sources. --
Albert Einstein 😊

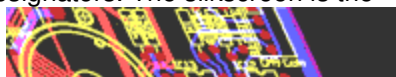
We put a bit of effort into the PCB design - it is somewhere where quality of design can make a difference. These days most circuit boards are automatically assembled and tested - but you will still have people working on your circuit boards, and having a quality design can make all the difference between a product that is pleasant to work on and one that they will hate. Machine test



points, for ATE (Automated Test Equipment) bed of nails testing, just need to be pads or lands. Some test points, however, are not really intended for ATE but rather for engineering test or modification - we put labelled test points on for these, as illustrated.

We put a complete silkscreen identification overlay on most of our boards, and the quality of this is another way we can make a better product. The silkscreen contains important information that assists people to service and install the product. Aside from functionality (which is rather obvious) the overlay is the chief distinguishing feature between a **purpose-built** product, such as we deliver, and a **general-purpose** product such as you might buy from a company like Rabbit. We would label terminals as: "+12v red" "-pwr black" "solenoid+" "solenoid-" whereas competitors may often just give you: "A1 A2 A3 A4". They don't have much of a choice - they have no idea what you will attach to the terminals.

For dense surface mount boards, we often need to leave off component values, and sometimes need to omit the component designators. The silkscreen is the primary method for labelling connectors, replaceable parts, orientation, and even



installation notes (for instance "Remove J2 while replacing battery - observe battery polarity")

The PCB layout also needs to take account of [EMI and ESD compliance](#) and we have a separate page on some techniques that we use to get our PCBs to comply. We have a [tutorial on designing PCBs](#) - it steps through the CAD process from schematic to layout. We also support the Autotrax/Easytrax CAD package by providing [Updated video drivers](#). If you are intending to design PCBs yourself, we would recommend this very good [manual on PCB layout](#).

Most of the Etched PCB laminates we produce are:

Double sided Laminate

Two layers of copper tracks, one each side of the board

Plated Through Hole

PTH - each hole is copper plated providing a circuit between sides of the PCB

Fibreglass-resin laminate ([FR4](#))

Solder Mask Over Bare Copper

SMOBC - green insulating ink everywhere except connections

Component legend

Identifying lettering, component outlines and values in white or yellow ink

Tinned

Tinning is application of Solder to all exposed copper, increasing solderability

Hot oil levelled

Hot oil - or hot air - levelling makes the tinning flat, so that surface mount components can be positioned reliably.

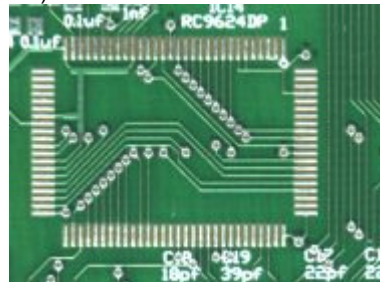
Different methods of PCB construction:

Conventional

A rigid PCB (usually of thickness 1.6mm), with wire-leaded components mounted on only one side of the PCB, with all the leads through holes, soldered and clipped. Conventional circuitry is generally easier to debug and repair than Surface mount.

Surface Mount Technology (SMT) or devices (SMD)

A PCB with tag-leaded components soldered flush to PCB pads. Holes are still needed on the PCB, but not where the component leads are attached. Surface mount circuitry is generally smaller than conventional. Surface mount is generally more suited to automated assembly than conventional.



Surface mount & conventional mix

In practice, most boards are a mix of surface mount and conventional components. This can have its disadvantages as the two technologies require different methods of insertion and soldering.

Double sided Laminate

A bare PCB laminate having tracks on both sides, normally with PTH holes connecting circuitry on the two sides together.

Double sided Component Assembly

Mounting components on both sides of the PCB. Normally only surface mount circuitry would be mounted on both sides of a PCB.

Multi-layer

A PCB Laminate may be manufactured with more than two layers of copper tracks by using a sandwich construction. The cost of the laminate reflects the number of layers. The extra layers may be used to route more complicated circuitry, and/or distribute the power supply more effectively.

Gold plated

Certain areas on a PCB may be gold plated for use as contact pads. Unless the whole PCB is gold plated before etching, this technique is limited in its application, normally, to pads on the edge of a PCB, as an electrolytic plating bar must be attached to the pads, and then removed part way through the PCB manufacturing process.

Flexible PCB

A technique used extensively with membrane keyboards, combination connector/circuit boards, and circuit boards to fit in awkward shapes - e.g. cameras.

Chip On Board (COB)

Where the IC die is attached direct to a PCB, and bond out wires from the IC connect directly to PCB lands. The chip is then covered with a black blob of epoxy. A technique used mostly with very high volume, cost sensitive applications, e.g. musical greeting cards.

Phenolic PCB

As distinct from Fibreglass, Phenolic is a cheaper PCB laminate material.

Daughterboard

A circuit board mounted to another circuit board - such as a plug in card.